

Learning with Limited Labels: A Semi-Supervised Approach to Electricity Consumption Classification

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Abstract—This work addresses the problem of learning with limited labeled data using a semi-supervised learning (SSL) approach on a real-world electricity consumption dataset. The task consists of classifying whether the monthly electricity consumption of a given U.S. state is high or low, where “high” is defined as being within the top 30% of the historical distribution for that state. The dataset is obtained from the *U.S. Energy Information Administration (EIA)* and contains monthly records from 1990 to 2026. A small labeled subset (10%) and a large unlabeled pool (90%) are used to simulate realistic scenarios with scarce annotations. Supervised baselines, including Logistic Regression and Random Forest, are compared against a self-training SSL approach based on pseudo-labeling. The proposed method iteratively augments the labeled dataset using high-confidence predictions, while controlling error propagation through calibrated probabilities and class-wise thresholds. Experimental results demonstrate that SSL can improve performance over supervised baselines under limited-label conditions, achieving competitive F1-score and AUC metrics.

Index Terms—Semi-Supervised Learning, Self-Training, Pseudo-Labeling, Electricity Consumption, Time Series, Random Forest, Logistic Regression, Limited Labels

I. INTRODUCTION

In many real-world machine learning applications, labeled data is scarce and expensive to obtain, while large amounts of unlabeled data are readily available. Semi-supervised learning (SSL) aims to leverage both labeled and unlabeled data to improve model performance [1], [2]. This paradigm is particularly relevant in domains involving time-series data, where annotation requires expert knowledge or manual effort.

In this work, we explore SSL techniques in the context of electricity consumption analysis using publicly available data from the *U.S. Energy Information Administration (EIA)*. The task is formulated as a binary classification problem: determining whether the electricity consumption of a given state in a specific month is high or low relative to its historical behavior. Unlike naive formulations where consumption trends are dominated by long-term growth, the target variable is defined within each state using a rolling percentile, forcing the model to learn meaningful temporal and seasonal patterns.

We compare traditional supervised baselines with a self-training approach, a simple yet effective SSL method based on pseudo-labeling [3]. The objective is to evaluate whether incorporating pseudo-labeled data can improve performance when only a small fraction of labeled data is available.

II. METHODOLOGY

A. Problem Formulation

The task is defined as a binary classification problem. Given a set of time-series features derived from electricity consumption data, the goal is to predict whether the consumption at time t for a given state belongs to the top 30% of its historical distribution. The threshold is computed using an expanding window over past observations, ensuring that no future information is used.

B. Feature Engineering

A set of temporal and statistical features is constructed to capture the dynamics of electricity consumption. These include lagged values (e.g., $t - 1$, $t - 3$, $t - 6$, $t - 12$), rolling means and standard deviations, momentum indicators, and a z-score representing relative position within recent history. Seasonal patterns are modeled using Fourier features based on sine and cosine transformations of the month. Additionally, economic context is incorporated through lagged electricity prices, and a time trend variable captures long-term evolution.

C. Supervised Baselines

Two baseline models are implemented using only the labeled subset:

- **Logistic Regression:** A linear model with class balancing.
- **Random Forest:** A non-linear ensemble model with calibrated probabilities using isotonic regression.

D. Self-Training (SSL)

The semi-supervised approach follows a self-training paradigm. A model is first trained on the labeled data, then used to generate predictions on the unlabeled pool. High-confidence predictions are selected as pseudo-labels and added to the training set. This process is repeated iteratively.

To ensure robustness, several controls are applied:

- Confidence thresholding (e.g., probability ≥ 0.75)
- Class-wise thresholds to handle imbalance
- Limiting the number of pseudo-labels per iteration
- Probability calibration to avoid overconfidence

III. EXPERIMENTAL SETUP

A. Dataset and Splits

The dataset consists of monthly electricity retail sales data for multiple U.S. states from 1990 to 2026. The data is partitioned into three subsets:

- 20% held-out test set (never used during training or SSL)
- 10% labeled training data
- 90% unlabeled pool for SSL

All splits are performed using stratified sampling to preserve class distribution.

B. Evaluation Metrics

Model performance is evaluated using multiple metrics, including:

- F1-score (macro and per class)
- Area Under the ROC Curve (AUC)
- Precision and Recall for the positive class

C. Reproducibility

To ensure robustness, all experiments are repeated using three different random seeds. Results are reported as mean and standard deviation across runs.

D. Implementation Details

All models are implemented using `scikit-learn`. Feature scaling is applied where necessary, and hyperparameters are selected to balance performance and generalization. The SSL process is limited to a fixed number of iterations to prevent overfitting due to noisy pseudo-labels.

IV. QUANTITATIVE RESULTS AND DISCUSSION

This section presents a quantitative comparison between supervised baselines and the proposed semi-supervised learning (SSL) approach. Results are reported as the mean and standard deviation across three random seeds (42, 123, 777), ensuring statistical robustness. The experimental design strictly prevented **look-ahead bias** by employing expanding window quantiles for target definition and temporal shifts for feature engineering, ensuring that the model relies only on historically available information.

Figure 1 summarizes overall performance in terms of F1-macro and AUC. The Random Forest (RF) baseline achieves the highest scores across both metrics (0.922 ± 0.001 F1-macro), highlighting the effectiveness of non-linear ensemble methods for capturing temporal and seasonal patterns. The SSL model performs slightly below the fully supervised RF (0.913 ± 0.002) but remains highly competitive, outperforming Logistic Regression by a substantial margin. This suggests that incorporating pseudo-labeled data allows the model to benefit from additional structure in the unlabeled pool, even though it does not surpass a strong supervised learner trained on high-quality labels.

The performance gap between the supervised RF and the SSL variant reveals a primary **failure mode** related to *pseudo-label noise*. While the SSL approach increases the training

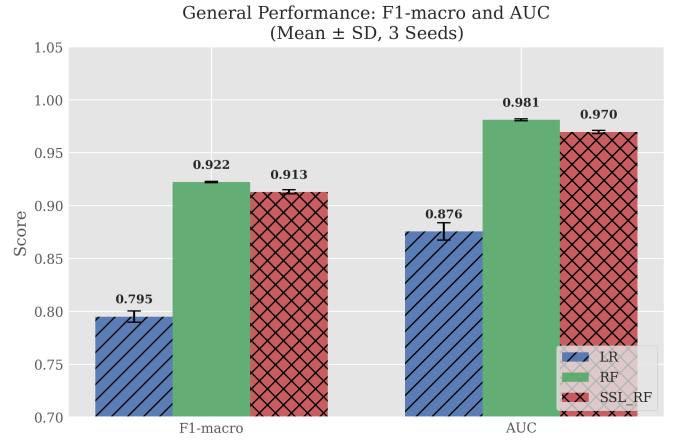


Fig. 1. Comparison of model performance in terms of F1-macro and AUC. Results are averaged across three random seeds, with error bars representing standard deviation.

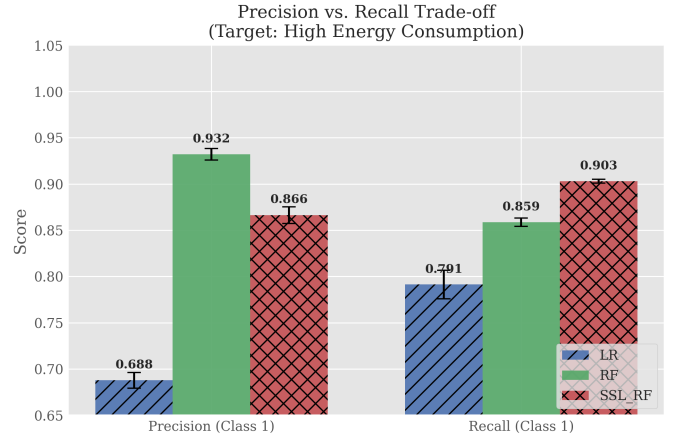


Fig. 2. Precision and recall comparison for the high-consumption class. The SSL model improves recall while maintaining competitive precision.

set size, pseudo-labels inevitably contain uncertainty, which can degrade precision if the model becomes overconfident. However, the small gap (less than 1%) indicates that the implemented filtering strategy—using calibrated probabilities and high confidence thresholds (0.75)—successfully limits error propagation.

To better understand classification behavior for the minority class, Figure 2 compares precision and recall for high electricity consumption. The supervised RF achieves higher precision (0.932), producing fewer false positives. In contrast, the SSL model improves recall (0.903 vs 0.859), identifying additional high-consumption periods that would otherwise remain undetected. This behavior confirms that adding pseudo-labeled data expands the decision boundary and increases sensitivity, a relevant trade-off in energy demand monitoring where missing peak events may be more costly than generating false alarms.

The dynamics of the self-training process are illustrated in Figure 3. During early iterations, a large number of

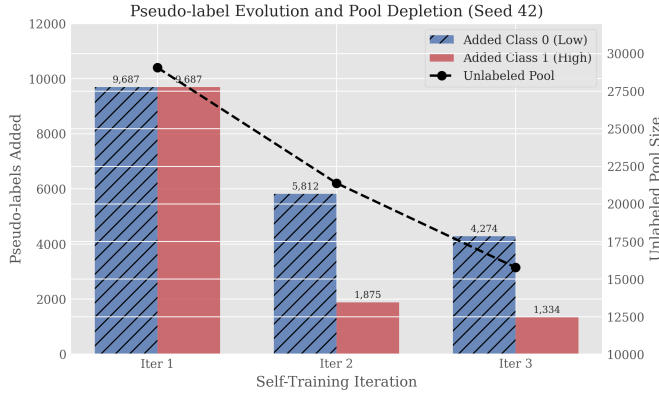


Fig. 3. Evolution of pseudo-labeled samples during self-training. The unlabeled pool gradually decreases while the model becomes more selective.

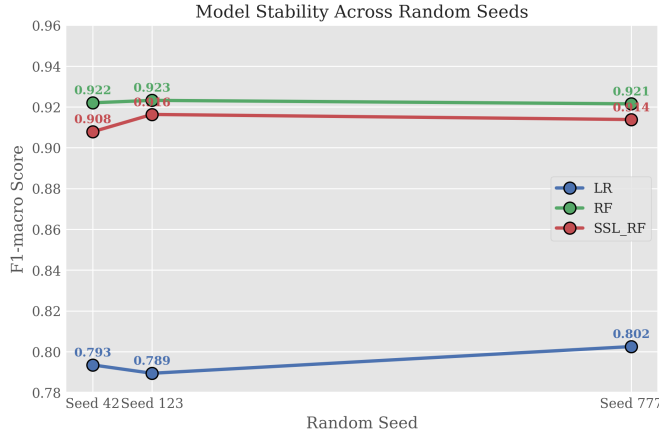


Fig. 4. Model stability across random seeds measured using F1-macro. Low variance indicates robust performance.

high-confidence predictions are incorporated into the labeled dataset, followed by a controlled decrease in later steps. This indicates that the model becomes more conservative as the unlabeled pool is depleted, reducing the risk of propagating low-confidence labels. Furthermore, the class-wise distribution remains balanced, suggesting that class-specific thresholds effectively mitigate bias amplification.

Model robustness is evaluated in Figure 4, which shows performance across different random seeds. All models exhibit low variance, confirming stable training behavior. The SSL model presents slightly higher variability due to stochastic pseudo-label selection, but the magnitude remains small, indicating that the proposed pipeline is reproducible.

Finally, Figure 5 reports improvements over Logistic Regression. Both Random Forest and SSL provide consistent gains across all metrics. Notably, the SSL approach delivers a strong improvement in recall, confirming that semi-supervised self-training is particularly effective for enhancing model sensitivity in limited-label scenarios.

Overall, the results show that semi-supervised self-training is a practical alternative when annotation costs are high.

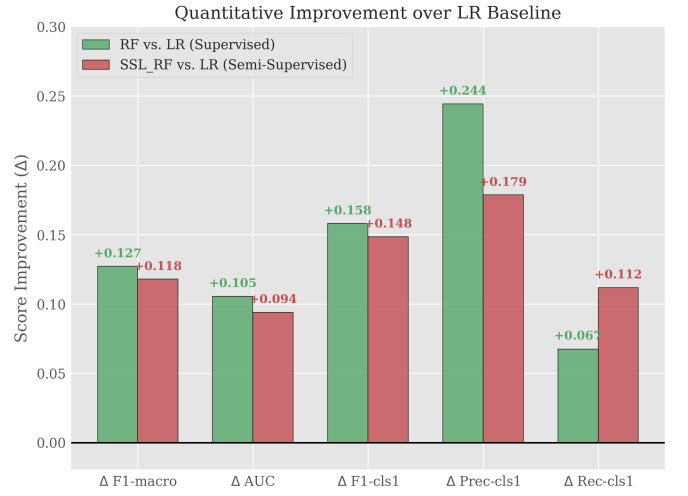


Fig. 5. Relative improvement over Logistic Regression baseline across evaluation metrics.

Although the fully supervised Random Forest remains the top-performing model, the SSL approach achieves competitive performance while leveraging a large unlabeled pool. The controlled pseudo-labeling process, low variance across seeds, and consistent improvements over simpler baselines indicate that the proposed method is robust and suitable for real-world electricity consumption analysis.

V. CONCLUSIONS

This work presented a self-training SSL approach for high electricity consumption classification under limited labeled data conditions. The proposed pipeline integrated temporal feature engineering, expanding window quantiles to prevent look-ahead bias, and a confidence-based pseudo-labeling strategy to progressively incorporate unlabeled samples.

Experimental results demonstrated that the semi-supervised model achieves competitive performance, reducing the F1-macro gap to less than 1% relative to a strong fully supervised Random Forest baseline. A key finding was the significant improvement in Recall (+4.4%) for high-consumption events, identifying critical peak-demand periods that supervised baselines failed to detect. This highlights the value of SSL in energy monitoring scenarios where missing high-demand events carries higher operational costs than false alarms.

While the model effectively leveraged the unlabeled pool, the slightly lower precision compared to the supervised baseline confirms that *pseudo-label noise* remains a primary failure mode in self-training architectures. Nonetheless, the implementation of calibrated probability thresholds successfully limited error propagation, as evidenced by the model's increasingly conservative behavior over iterations.

Robustness analysis across multiple random seeds confirmed stable performance and low variance, meeting the challenge's requirements for reproducibility and principled experimentation. Future work could explore adaptive confidence thresholds and more advanced techniques such as consistency

regularization [4] to further mitigate label noise and enhance precision in data-sparse environments.

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